

Ketan Kumar Bagewadi

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PROFESSIONAL SUMMARY

Software Engineer skilled in Python, Machine Learning, LLM, NLP, RAG, Backend Development, Docker Compose, Kubernetes, Docker and Jenkins. Passionate about building scalable AI applications and backend systems. Seeking AI/ML or DevOps roles.

EXPERIENCE

NetSense CyberSecurity Pvt. Ltd.

Mar 2025 - Present

- Software Developer working on **Python, Machine Learning** and **Deep Learning** to detect and flag **phishing and malicious websites**. Achieved 95%-97% accuracy in detecting true phishing websites on live traffic.
- Designed and deployed dual production website classification for **40+ categories** using **Llama 3.1 8B** and 250-thread **Selenium** pipeline and also implemented using the **Sentence-t5-xxl NLP** model with cosine-similarity and zero-shot classification.
- Deployed **CRM** system using **Python, Django, Django REST Framework, JavaScript, Ajax, Linux, Nginx, Azure Directory, DigitalOcean Droplet** and **PostgreSQL**.
- Optimised **CRM Web App** to query from and sync with **2+ Database servers**. Accomplished **30%** faster read and write operations.
- Built a **LLM-based network anomaly detection** system using DNS and NetFlow data with rule-based +LLM (**Meta-Llama-3.1 8B**) to detect cyber threats.

PROJECTS

DeepFake Detection | Python, Google Colab, PyTorch, HTML, CSS, JavaScript

[Github Link](#)

- Developed a **deepfake detection** project utilizing **deep learning** techniques. Employed the **FaceForensics++** and **Celeb-DF** database to train a robust **machine learning** model.
- Implemented an architecture combining **ResNeXt** and **LSTM** models for accurate identification and classification of deepfake videos.

LLM-Powered RAG ChatBot | Python, LangChain, OpenAI, Ollama, LLM, RAG

[Github Link](#)

- Developed an AI chatbot using **GPT-4o-mini**, **Llama 3** fallback, and **LangChain** to deliver reliable, real-time conversational responses.
- Built a **RAG-based chatbot** using **ChromaDB**, **document chunking**, **embeddings**, and **semantic retrieval** for context-aware responses and improved answer accuracy. **Containerized** the frontend and backend using **Docker**, **orchestrated** services with **Docker Compose**, and deployed the RAG application using **Kubernetes**.

AI ChatBot Android App | Android Studio, Java, Firebase, Google PaLM-2 API

[Github Link](#)

- Developed an **Android application** proficient in processing text input and generating responses for a wide array of questions sourced from the **Google PaLM-2 API**.
- Engineered an application with capabilities of **ChatGPT** like conversational capabilities.

TECHNICAL SKILLS

Languages: Python, C | **Frameworks & Libraries:** PyTorch, Scikit-learn, Pandas, Django REST Framework, Bootstrap, TailwindCSS | **Databases:** PostgreSQL, MySQL | **DevOps:** Docker, Kubernetes, Jenkins, Nginx, DigitalOcean | **Cloud:** AWS, Azure | **Version Control:** Git, GitHub | **Other:** Unix/Linux, Machine Learning, Deep Learning, NLP, Data Analysis, Web Scraping, LLM, Bert, RAG, Llama, Prompt Engineering, AI Tools (ChatGPT, Claude AI, Grok AI, Google AI, Cursor, Mistral AI)

EDUCATION

Bachelor of Engineering in Computer Science & Engineering

KLE Dr. M.S. Sheshgiri College of Engineering and Technology, Belgaum, Karnataka

June 2024

CERTIFICATIONS

AWS Cloud Practitioner Essentials Course (**Amazon**), Introduction to Python (**Sololearn, Infosys Springboard**), Introduction and Intermediate SQL (**Great Learning**), Generative AI (**GeeksforGeeks**), Introduction to Data Science (**Cisco**)